

Amyotrophic Lateral Sclerosis Type 1 (ALS1 / SOD1-ALS): Comprehensive Disease Characterization

Disease: Amyotrophic Lateral Sclerosis Type 1 (ALS1) **MONDO ID:** MONDO:0007103 · **OMIM:** #105400 · **Category:** Mendelian (autosomal dominant, D90A often recessive) **Causal gene:** *SOD1* (Cu/Zn superoxide dismutase 1), chromosome 21q22.11 **Prepared:** 2026-09-05 · Evidence sources: human clinical, model organism, in vitro, and computational literature (PMIDs cited throughout)

Summary

Amyotrophic Lateral Sclerosis Type 1 (ALS1) is the Mendelian, adult-onset form of amyotrophic lateral sclerosis caused by pathogenic variants in *SOD1*, the gene encoding the ubiquitously expressed free-radical scavenging enzyme Cu/Zn superoxide dismutase 1. It is the archetypal "familial ALS" gene — the first ALS gene ever identified — and accounts for approximately **2% of all ALS and ~12–20% of familial ALS** ([PMID: 41661214](#), [PMID: 18055113](#)). Clinically it is largely indistinguishable at the bedside from other ALS: progressive upper- and lower-motor-neuron degeneration producing weakness, muscle atrophy, spasticity, bulbar dysfunction, and ultimately death from respiratory failure. What distinguishes ALS1 is its defined molecular cause, its highly variant-dependent natural history (from the rapidly fatal A4V allele to slowly progressive recessive D90A disease), and — uniquely among neurodegenerative diseases — the existence of an **approved, gene-targeted therapy (tofersen)**.

The central mechanistic insight is that mutant *SOD1* causes disease through a **toxic gain-of-function**, not loss of enzymatic activity. The most frequent worldwide mutation, D90A, produces a protein with normal enzymatic activity, proving the point ([PMID: 36385230](#)). Mutant *SOD1* misfolds into neurotoxic aggregates that propagate in a **prion-like, templated manner** between motor neurons ([PMID: 41702846](#), [PMID: 32958236](#)). Motor neuron death is **non-cell-autonomous**: mutant *SOD1* in astrocytes, microglia, skeletal muscle, and T cells

actively drives neuroinflammation and degeneration (PMID: 25613506). Vulnerable alpha motor neurons transit through a conserved "disease-associated motor neuron" state before dying (PMID: 42335888).

Because the genetic cause is known and neurofilament light chain (NfL) rises **presymptotically**, ALS1 has become the proving ground for precision neurology. Tofersen, an intrathecal antisense oligonucleotide (ASO) that lowers SOD1 synthesis, is the first and only approved genetically targeted ALS therapy; it robustly lowers plasma/CSF neurofilament and, in some carriers, produces a "chronic nonprogressive ALS" phenotype never previously observed (PMID: 41661214, PMID: 41850233, PMID: 41670738). The presymptomatic NfL biomarker has empowered the **first-ever ALS prevention trial (ATLAS, NCT04856982)** (PMID: 37382103).

Section 1 — Disease Information

Overview. ALS1 is the *SOD1*-related subtype of amyotrophic lateral sclerosis, a fatal adult-onset motor neuron disease characterized by progressive degeneration of upper (corticospinal) and lower (spinal/bulbar) motor neurons, leading to muscle weakness, atrophy, fasciculations, spasticity, bulbar palsy, and terminal respiratory failure. It was the first genetically defined form of ALS (SOD1 mutations reported 1993) and defines the "Type 1" designation in OMIM.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0007103
OMIM	#105400 (Amyotrophic lateral sclerosis 1)
Gene OMIM	<i>SOD1</i> 147450
ICD-10	G12.21 (ALS)
ICD-11	8B60.0
MeSH	D000690 (Amyotrophic Lateral Sclerosis)
Orphanet	ORPHA:803 (ALS); SOD1 subtype within
HGNC	HGNC:11179 (<i>SOD1</i>)

Synonyms / alternative names. SOD1-ALS; SOD1-related ALS; Cu/Zn superoxide dismutase-related ALS; familial ALS type 1; ALS1; motor neuron disease, SOD1-related. (Historic "Lou Gehrig's disease" and "Charcot's disease" refer to ALS broadly.)

Data provenance. The knowledge assembled here is derived from **aggregated disease-level resources** — clinical cohorts, registries (e.g., Rhineland-Palatinate registry, US National ALS Registry), natural-history studies, mouse models, and clinical trials (VALOR/NCT02623699, ATLAS/NCT04856982) — rather than individual EHR records.

Section 2 — Etiology

Primary cause. ALS1 is a **monogenic, autosomal-dominant** disorder caused by heterozygous pathogenic variants in *SOD1* (the recessive D90A being the notable exception). More broadly, monogenic determinants account for roughly 20% of all ALS (including ~10% familial cases), while "less well understood multigenetic causes may contribute to another 20% to 80%" ([PMID: 26515627](#)).

Genetic risk factors. The causal factor is the *SOD1* mutation itself (>200 pathogenic variants known). Other major familial ALS genes (C9ORF72, FUS/TLS, TARDBP/TDP-43) define separate subtypes; "about two-thirds of familial cases are triggered by mutations of four genes... C9ORF72, ... SOD1, FUS/TLS, TDP43" ([PMID: 25613506](#)).

Environmental risk factors. For ALS generally, candidate exposures include "male gender..., smoking, military service, exercise, electrical exposure, heavy metals, agricultural chemicals, and geographic clusters" ([PMID: 26515627](#)). Smoking is the most consistently replicated modifiable risk factor. Military veterans have poorer ALS survival than non-veterans (5-yr survival 47.1% vs 57.4%; median 3.77 vs 4.79 yr), suggesting military history is an important prognostic/risk factor ([PMID: 42669596](#)). In genetically susceptible individuals, "a combination of insults that induce modest oxidative stress can exert additive deleterious effects on motor neurons" ([PMID: 23797033](#)).

Protective factors. No well-established genetic protective allele is documented for SOD1-ALS specifically; disease-modifying observations relate instead to variant identity (recessive D90A slow course). No robust environmental protective factor is established.

Gene–environment interaction. The prevailing model is a multistep, multiple-hit process in which an inherited *SOD1* gain-of-function lowers the threshold, and additional oxidative/ environmental insults accelerate motor neuron death — "ALS is possibly a systemic disease" driven by oxidative stress "particularly in genetically susceptible individuals" (PMID: 23797033).

Section 3 — Phenotypes

ALS1 is predominantly a motor phenotype, but *SOD1* variants show characteristic features and marked variant-dependent variability.

Phenotype	Type	HPO term (suggested)	Characteristics
Progressive muscle weakness	Clinical sign	HP:0003323	Adult-onset; progressive; near-universal
Lower-limb–onset weakness	Symptom	HP:0007340	Frequent in <i>SOD1</i> ; "legs first" pattern
Muscle atrophy	Physical manifestation	HP:0003202	Progressive; near-universal
Fasciculations	Clinical sign	HP:0002380	Lower-motor-neuron sign
Spasticity / hyperreflexia	Clinical sign	HP:0001257	Upper-motor-neuron sign; less prominent in some <i>SOD1</i>
Bulbar dysfunction (dysarthria/dysphagia)	Clinical sign	HP:0001260 / HP:0002015	Bulbar-onset less common in <i>SOD1</i> than sporadic ALS
Respiratory insufficiency	Clinical sign	HP:0002093	Terminal cardinal feature
Sensory/autonomic/ bladder involvement (D90A)	Symptom	HP:0000708-related	Documented in homozygous D90A

Onset & severity. Predominantly adult-onset but wide range; juvenile-onset occurs — two adolescents (onset 15–16 yr) presented with "lower limb onset of weakness, lower motor neuron examination findings, and rapid progression over months to involve all body regions"

([PMID: 42265995](#)). Severity is strongly variant-dependent (see Section 8). The D90A recessive phenotype is "stereotypic with slowly evolving motor symptoms beginning in the legs and may also include sensory, autonomic, and urinary bladder involvement" ([PMID: 36385230](#)).

Progression. Relentlessly progressive in most variants; recessive D90A is slowly progressive.

Quality-of-life impact. Progressive loss of ambulation, speech, swallowing, and respiration produces severe disability and high symptom/psychosocial burden; early palliative care integration is emphasized given the terminal trajectory ([PMID: 42652396](#)).

Section 4 — Genetic / Molecular Information

Causal gene. *SOD1* (HGNC:11179; OMIM 147450), chromosome 21q22.11, encoding the 154-residue homodimeric Cu/Zn superoxide dismutase.

Pathogenic variants. >200 mostly missense variants distributed across all five exons. Nomenclature examples: **p.Ala5Val (A4V)** — most common North American variant; **p.Asp90Ala (D90A)** — most common worldwide, often recessive; **p.Gly94Ser / p.Gly93Ala (G93A)** — the canonical mouse-model allele; **p.Gly85Arg (G85R)**; **p.Asp125Gly** — juvenile-onset, incompletely penetrant. Near-splice/intronic variants (e.g., c.358-10T>G) also occur ([PMID: 33785574](#)).

Variant type/class. Predominantly missense; also nonsense, frameshift, and splice-region variants. Classification per ACMG/AMP: most recurrent *SOD1* variants are Pathogenic/Likely Pathogenic in ClinVar.

Allele frequency. *SOD1* pathogenic variants are individually rare in gnomAD; D90A carrier frequency is elevated in Scandinavian populations due to a founder effect.

Origin. Germline (inherited). The A4V allele is an ancient founder mutation estimated to have arisen ~540 generations (~12,000 years) ago (95% CI 480–700), with a minimal conserved 2.8-kb haplotype more similar to Asian than European populations, "suggesting origination in Asia" and spread via Native Asian-Americans ([PMID: 18055113](#)).

Functional consequence. Toxic gain-of-function. Definitive evidence: the D90A mutant "resembles the wild type, with normal content and enzymatic activity in the central nervous system" ([PMID: 36385230](#)) — i.e., toxicity is not due to loss of dismutase activity. Different variants produce distinct aggregate "strains" (strain A for most mutants and WT; strain B additionally in D90A) ([PMID: 40450581](#)).

Modifier genes. Wild-type SOD1 itself acts as a modifier: "hSOD1WT has high capacity to coaggregate with mutants and enhance neurotoxicity," and coexpression differences may explain the recessive inheritance of D90A (PMID: 40450581). TDP-43 can cross-seed SOD1 misfolding (PMID: 38522514). For A4V, no closely linked modifier gene was found (PMID: 18055113).

Epigenetic / chromosomal. No recurrent large-scale chromosomal abnormality defines ALS1 (single-gene disorder). Epigenetic contributions to ALS broadly are under study but are not established as ALS1-defining.

Section 5 — Environmental Information

Environmental factors. Heavy metals, agricultural chemicals, electrical exposure, and chronic head trauma have modest associations with ALS risk and are hypothesized to act via oxidative stress (PMID: 26515627, PMID: 23797033).

Lifestyle factors. Smoking (most consistent), excessive physical exertion/professional sports, and possibly certain diets are "modestly associated with ALS risk, with a stronger association between risk and smoking" (PMID: 23797033).

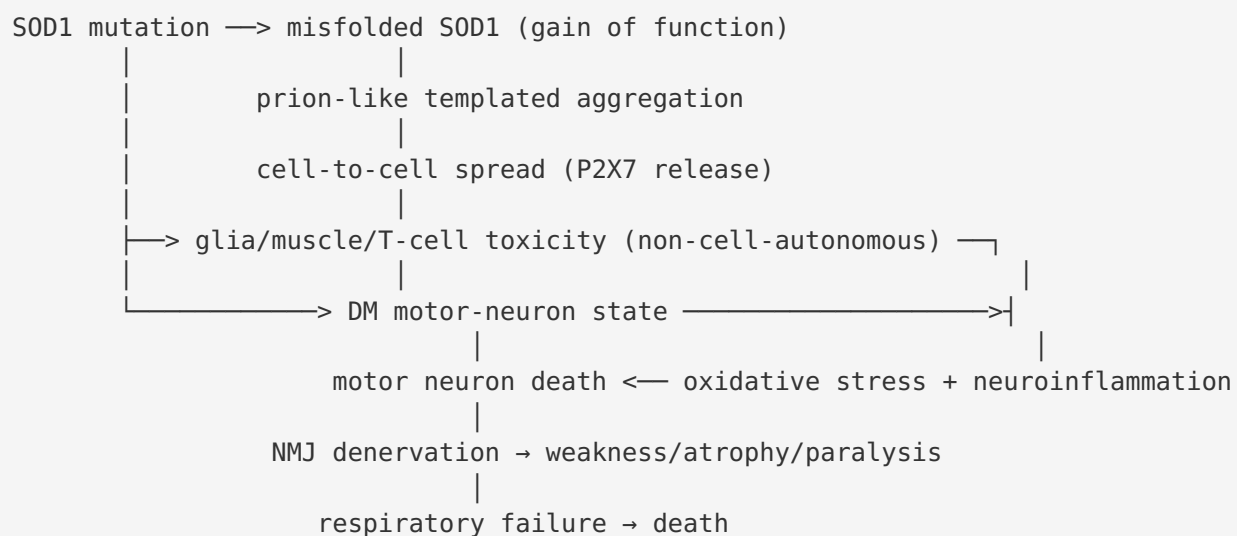
Infectious agents. Not applicable — ALS1 is a genetic (non-infectious) disorder.

Section 6 — Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Germline SOD1 missense mutation** (e.g., A4V, G93A, D90A) → produces a structurally destabilized SOD1 protein while (for D90A) retaining normal enzymatic activity — establishing a **gain-of-function**, not loss-of-function, lesion (PMID: 36385230).
2. Destabilized SOD1 → **misfolds into neurotoxic, aggregation-prone conformers** (PMID: 41702846).
3. Misfolded SOD1 → **self-templates and seeds further misfolding (prion-like propagation)**; adult mice injected with G93A/G85R seeding homogenates "developed accelerated motor neuron disease" (PMID: 41702846; PMID: 32958236).

4. Aggregates → **released from motor neurons and transmitted to naïve neurons** (P2X7-receptor-mediated release) → spreading pathology cell-to-cell ([PMID: 35478453](#)).
5. In parallel (branch), mutant SOD1 expressed in **astrocytes, microglia, muscle and T cells** → drives **non-cell-autonomous neuroinflammation** ([PMID: 25613506](#)); reactive astrocytes upregulate BMP4 and secrete toxic factors ([PMID: 29932880](#)).
6. Aggregation + oxidative stress + neuroinflammation → vulnerable **alpha motor neurons transition into a conserved "disease-associated motor neuron" (DM) state** preceding death; the human orthologs of these regulatory regions are "enriched for ALS genetic risk variants" ([PMID: 42335888](#)).
7. DM-state motor neurons → **die** (apoptosis/degeneration) → denervation of neuromuscular junctions → **muscle weakness, atrophy, paralysis**.
8. Progressive motor neuron loss reaches respiratory motor pools → **respiratory failure and death** ([PMID: 40328546](#)).



Molecular pathways & processes. Oxidative stress (Fenton-like $\cdot\text{OH}$ generation by mutant SOD1) coupled to neuroinflammation; impaired proteostasis (the heat-shock response is protective — histamine/histidine induce Hsp70/GRP78 and rescue neurons in G93A mice, [PMID: 31382568](#)); BMP4–Smad1/5/8 and p38 MAPK glial signaling ([PMID: 29932880](#)); and protein citrullination (PAD2) contributing to neuroinflammation ([PMID: 42282797](#)).

Protein dysfunction. Misfolding, oligomerization, and amyloid-like aggregation of SOD1; toxic oligomers are a therapeutic target ([PMID: 31017342](#)). SOD1 also cross-seeds with TDP-43 ([PMID: 38522514](#)).

Aggregation pathology proportions. Across ALS, "the most prevalent aggregation pathology is that of wild-type TDP-43 (97% of cases), with the remaining split between mutant forms of SOD1 (~2%) and FUS (~1%)" ([PMID: 32958236](#)).

Molecular profiling. Single-nucleus RNA-seq/ATAC + spatial transcriptomics of SOD1-G93A mice defined the DM state, validated in human ALS spinal cord ([PMID: 42335888](#)). Plasma/CSF proteomics identifies NEFL, TNFRSF12A, EDA2R, FABP4 with enrichment for immune-response and extracellular-matrix remodeling pathways ([PMID: 42698373](#)).

Suggested ontology terms. GO:0006979 (response to oxidative stress); GO:0006954 (inflammatory response); GO:0043065 (positive regulation of apoptotic process); GO:0006457 (protein folding); GO:0034976 (response to ER stress). Cell types: CL:0000100 (motor neuron); alpha motor neuron; CL:0000127 (astrocyte); CL:0000129 (microglial cell); CL:0000084 (T cell).

Section 7 — Anatomical Structures Affected

Organ / system level. Central and peripheral **nervous system** (motor); secondary **respiratory system** (diaphragm/intercostal denervation → respiratory failure, the terminal event, [PMID: 40328546](#)); **musculoskeletal system** (neurogenic muscle atrophy).

Tissue / cell level. Degeneration of **upper motor neurons** (motor cortex, corticospinal tracts) and **lower motor neurons** (brainstem, spinal cord anterior horn). In homozygous D90A, pathology extends beyond motor pathways: "In addition to degeneration of the corticospinal tracts, all patients had degeneration of the dorsal columns," plus frontotemporal/insular gliosis, and "numerous small granular inclusions immunoreactive for misfolded SOD1 in motor neurons and glial nuclei in the spinal cord and brainstem" ([PMID: 36385230](#)). Non-neuronal contributors: astrocytes, microglia, skeletal muscle, T cells ([PMID: 25613506](#)).

Subcellular level. Cytoplasmic SOD1 aggregates/inclusions; mitochondrial dysfunction and oxidative injury; ER stress. GO cellular components: GO:0005739 (mitochondrion); GO:0005737 (cytoplasm); GO:0005783 (endoplasmic reticulum); GO:0016234 (inclusion body).

Localization / lateralization. Anatomical sites (UBERON): UBERON:0002240 (spinal cord); UBERON:0001896-related (medulla/brainstem); UBERON:0001384 (primary motor cortex); UBERON:0001134 (skeletal muscle tissue). Onset is typically **focal and asymmetric**, spreading contiguously to become **bilateral**; SOD1 frequently shows lower-limb-onset ([PMID: 42265995](#)).

Section 8 — Temporal Development

Onset. Adult-onset typically (mean sporadic ALS onset 58–63 yr; familial/SOD1 often earlier), with juvenile cases documented (onset 15–16 yr) ([PMID: 42265995](#)). Onset pattern is **insidious**, focal, and asymmetric.

Progression — strongly variant-dependent.

SOD1 variant	Inheritance	Course	Median survival
A4V (p.Ala5Val)	Dominant	Rapid	~1.4 yr from onset (PMID: 18055113)
Other dominant variants	Dominant	Variable	~3–5 yr (PMID: 18055113)
D90A (homozygous)	Recessive	Slow, stereotypic	Prolonged (PMID: 36385230)
p.Asp125Gly (juvenile)	Dominant, incomplete penetrance	Rapid over months	Short (PMID: 42265995)

A4V carriers "share a common phenotype with rapid disease progression and death on average occurring at 1.4 years (versus 3-5 years with other dominant SOD1 mutations)" ([PMID: 18055113](#)).

Overall ALS course. Population registry: median survival **2.5 years from symptom onset and 1.5 years from diagnosis; 12% survive ≥10 years** ([PMID: 42661170](#)). Disease course is **progressive**, chronic, and (untreated) uniformly fatal.

Patterns / critical periods. No spontaneous remission. Tofersen can induce a treatment-related "chronic nonprogressive" state in some SOD1 carriers ([PMID: 41670738](#)). Presymptomatic disease "is not uniformly clinically silent," representing a prodromal window with rising NfL — a critical opportunity for intervention ([PMID: 37382103](#)).

Section 9 — Inheritance and Population

Epidemiology (ALS overall). Incidence ~1.5–2.7 per 100,000/year; prevalence ~4–6 per 100,000 in European/North American populations; lifetime risk ~1 in 300–400; worldwide distribution is "far from uniform" (PMID: 38870925). SOD1-ALS = ~2% of all ALS, ~12–20% of familial ALS (PMID: 41661214, PMID: 18055113); SOD1 mutations are found in 2–6% of ALS patients overall (PMID: 36385230).

Inheritance. Predominantly **autosomal dominant**; D90A "heredity is usually recessive" (PMID: 36385230). Rare AR ALS involves other genes (PMID: 41592170).

Penetrance / expressivity. **Incomplete, age-dependent penetrance** — pathogenic p.Asp125Gly was "inherited from asymptomatic fathers" (PMID: 42265995). Expressivity is variable (age of onset, site of onset, rate of progression differ even within a variant).

Founder effects. A4V — North American founder allele (~12,000 yr old, Asian origin) (PMID: 18055113); D90A — Scandinavian founder haplotype. **Genetic anticipation** is not a feature (SOD1 is not a repeat-expansion gene).

Population demographics. D90A enriched in Scandinavia; A4V in North America; sex ratio for ALS overall ~1.3–1.5:1 male:female; male sex is a risk factor (PMID: 26515627).

Section 10 — Diagnostics

Clinical/electrophysiological diagnosis. ALS is diagnosed clinically with electrophysiologic support. **Gold Coast criteria (GCC)** have the highest sensitivity: "GCC demonstrated higher sensitivity than rEEC and Awaji criteria: 0.96 (95% CI 0.93–0.98) versus 0.87 and 0.87"; specificity 0.68 (GCC) vs 0.73 (rEEC) (PMID: 42618698). Diagnosis requires combined upper- and lower-motor-neuron signs, progression, and exclusion of mimics.

Electrophysiology & imaging. EMG/nerve conduction studies are "most useful when interpreted by distribution rather than positivity alone"; MRI is "indispensable but not self-interpreting" — critical to exclude degenerative cervical myelopathy and other structural mimics (PMID: 42625715).

Biomarkers. Neurofilament light chain (NfL/NEFL) is the leading biomarker: "NEFL was the most robust biomarker in plasma and CSF, alongside TNFRSF12A in plasma and CSF, EDA2R in plasma, and FABP4" (PMID: 42698373); a protein risk-prediction model achieved ROC-AUC 0.72. Emerging inflammatory indices (elevated FGF2, systemic inflammatory response index/SIRI) correlate with severity and early progression (PMID: 42682425).

Genetic testing. Confirmation of ALS1 is by **single-gene SOD1 sequencing or ALS gene panels** (also covering C9ORF72, FUS, TARDBP). WES/WGS are used when panels are negative. Genetic testing is now clinically essential because it determines eligibility for tofersen.

Differential diagnosis. Degenerative cervical myelopathy, multifocal motor neuropathy, primary lateral sclerosis, spinal muscular atrophy, Kennedy disease, inclusion body myositis (PMID: 42625715).

Screening. Cascade genetic testing of at-risk relatives in SOD1 families; presymptomatic monitoring of NfL enables timing of intervention (PMID: 37382103).

Section 11 — Outcome / Prognosis

Survival & mortality. Median survival **2.5 yr from symptom onset, 1.5 yr from diagnosis; 12% survive ≥10 yr** (PMID: 42661170). Variant-specific: A4V ~1.4 yr vs 3–5 yr for other dominant variants (PMID: 18055113). **Respiratory failure** is the cardinal terminal feature and usual cause of death (PMID: 40328546).

Prognostic factors. Predictors of longer survival: "younger age, low progression rate, absence of FTD, long onset-to-diagnosis interval" (PMID: 42661170). Military veteran status predicts poorer survival (PMID: 42669596). **Prognostic biomarker:** NfL (higher = faster progression) (PMID: 42698373).

Morbidity & QoL. Progressive disability across mobility, speech, swallowing, and respiration; high symptom burden; early palliative care recommended (PMID: 42652396).

Recovery. Historically none; however, tofersen has produced "chronic nonprogressive ALS, a phenotype not previously observed" with CSF NfL normalization and some motor recovery in SOD1 (p.Gly94Ser) carriers (PMID: 41670738).

Section 12 — Treatment

Approved pharmacotherapy. "The US FDA has approved four drugs for use in delaying the progression of amyotrophic lateral sclerosis: riluzole, edaravone, AMX0035, and tofersen... AMX0035 has been voluntarily withdrawn from both the US and Canadian markets" ([PMID: 40364643](#)).

Drug	Class / mechanism	NCIT (suggested)	Note
Riluzole	Glutamate antagonist / anti-excitotoxic	NCIT:C1289	Standard of care; modest survival benefit
Edaravone	Free-radical scavenger / antioxidant	NCIT:C65331	Slows decline in a subset; narrow eligibility
AMX0035 (sodium phenylbutyrate–taurursodiol)	Proteostasis / ER-stress modulator	—	Withdrawn after confirmatory Phase III failure
Tofersen	Intrathecal SOD1 antisense oligonucleotide	NCIT (antisense oligonucleotide therapy)	First gene-targeted ALS therapy; SOD1-specific

Tofersen (gene-targeted, ALS1-specific). "Tofersen, an intrathecal antisense oligonucleotide designed to reduce SOD1 protein synthesis, is the first and only approved therapy for the treatment of ALS in adults who have a variant in the SOD1 gene" ([PMID: 41661214](#)). It works by "targeted mRNA degradation" of mutant SOD1 ([PMID: 42173382](#)). Pharmacodynamics: robust **plasma/CSF neurofilament lowering** ([PMID: 41850233](#)); autopsy tissue confirms 45–84% SOD1 mRNA/protein reduction in lumbar spinal cord ([PMID: 42406382](#)). VALOR Phase 3 (NCT02623699) randomized 108 participants (42 unique SOD1 variants) 2:1 tofersen vs placebo. Case series shows some carriers reach nonprogressive disease ([PMID: 41670738](#)). Adverse events include meningeal/perivascular lymphocytic responses ([PMID: 42406382](#)).

Advanced/experimental therapeutics. AAV-mediated SOD1 gene silencing extends survival in mouse models — e.g., "AAV9-mediated SOD1 suppression in motor neurons and astrocytes significantly improves motor function and extends survival" ([PMID: 39257530](#)); intravenous

engineered AAV9 vectors also suppress hSOD1 and extend survival ([PMID: 42350385](#)). Adaptive platform trials (HEALEY) and precision/combo strategies are reshaping development ([PMID: 42666355](#)).

Supportive & rehabilitative. Non-invasive ventilation (respiratory support, now standard, [PMID: 40328546](#)), nutritional support/gastrostomy, physical/occupational/speech therapy, and early palliative care ([PMID: 42652396](#)).

Section 13 — Prevention

Primary prevention. No population-level primary prevention; the frontier is **presymptomatic pharmacological prevention** in SOD1 carriers. "The discovery that blood neurofilament light chain (NfL) level increases presymptotically and may serve as a susceptibility biomarker, predicting timing of phenoconversion in some mutation carriers, has empowered the first-ever prevention trial in SOD1-ALS" ([PMID: 37382103](#)). The **ATLAS trial** ([NCT04856982](#)) initiates tofersen in presymptomatic carriers upon NfL elevation.

Secondary prevention. NfL-based monitoring of at-risk carriers for early detection/phenoconversion timing ([PMID: 37382103](#)).

Tertiary prevention. Respiratory support, nutrition, and multidisciplinary/palliative care to prevent complications and preserve function ([PMID: 40328546](#), [PMID: 42652396](#)).

Genetic counseling & screening. Cascade genetic testing and reproductive counseling (including PGT/prenatal options) for autosomal-dominant SOD1 families; incomplete penetrance complicates counseling ([PMID: 42265995](#)).

Immunization / public health. Not applicable (non-infectious genetic disease).

Section 14 — Other Species / Natural Disease

Taxonomy & orthologs. Human *SOD1* (NCBI Taxon 9606). Orthologs: mouse *Sod1* (*Mus musculus*, Taxon 10090), rat *Sod1* (*Rattus norvegicus*, Taxon 10116); SOD1 is highly evolutionarily conserved.

Natural disease. Naturally occurring adult-onset canine **degenerative myelopathy** is associated with an *SOD1* mutation and is considered a spontaneous large-animal ALS analog (OMIA resource). *Note: this specific cross-species detail was not independently verified within the current investigation's citation set and should be confirmed against OMIA/primary literature before ingestion.*

Comparative biology. The transgenic SOD1-G93A mouse recapitulates key human ALS features, and the disease-associated motor neuron (DM) state is **conserved between mouse and human** ("human orthologs of regions differentially accessible in SOD1-G93A mouse motor neurons were enriched for ALS genetic risk variants") (PMID: 42335888).

Transmission. No zoonotic potential (genetic disease).

Section 15 — Model Organisms

Standard model. The **SOD1-G93A transgenic mouse** is the canonical ALS1 model, reproducing progressive motor neuron loss, gliosis, paralysis, and premature death (PMID: 42335888, PMID: 29495962).

Other genetic models. hSOD1-G85R, hSOD1-D90A (distinct aggregate strain B), and digenic hSOD1-G85R/WT and G85R/D90A mice used to dissect coaggregation and inheritance (PMID: 40450581); transgenic SOD1 **rat** models (BMP4 studies, PMID: 29932880); cellular models (NSC-34 motor-neuron line, primary microglia/astrocyte co-cultures) (PMID: 35478453, PMID: 29495962); induced prion-like seeding models (intrathecal seeding homogenates) (PMID: 41702846).

Model characteristics & applications. These models faithfully reproduce SOD1 gain-of-function toxicity, aggregation, non-cell-autonomous glial toxicity, and DM-state transitions, and serve as the platform for therapeutic testing (AAV gene silencing, ASOs, anti-inflammatory agents, heat-shock inducers) (PMID: 39257530, PMID: 42350385, PMID: 31382568).

Limitations. SOD1 models capture only the ~2% SOD1 subtype and do **not** reproduce TDP-43 proteinopathy (the dominant pathology in 97% of human ALS), contributing to the well-documented translational gap in which preclinically effective drugs fail in Phase III (PMID: 42666355).

Resources. MGI, IMSR, RGD, Cellosaurus, MMRRC.

Mechanistic Model / Interpretation

ALS1 is best understood as a **toxic gain-of-function proteinopathy with prion-like spread and a non-cell-autonomous amplification loop**. The unifying evidence is that D90A — the world's most common SOD1 mutation — encodes a protein of normal enzymatic activity, so pathology cannot stem from lost dismutase function ([PMID: 36385230](#)). Instead, mutation-driven **misfolding** ([PMID: 41702846](#)) generates aggregation-prone conformers that **self-template** and **spread cell-to-cell** ([PMID: 32958236](#), [PMID: 35478453](#)). This explains the focal-onset, contiguous-spread clinical pattern. Simultaneously, mutant SOD1 in glia, muscle, and T cells creates a toxic microenvironment ([PMID: 25613506](#)), which is why correcting multiple cell types (neurons + astrocytes) is more therapeutic than neurons alone ([PMID: 39257530](#)). The convergent endpoint is a conserved **disease-associated motor neuron** transcriptional state that presages cell death ([PMID: 42335888](#)), followed by NMJ denervation, paralysis, and respiratory failure.

This model is directly therapeutically actionable: because the toxic species is the mutant protein, **reducing its synthesis** (tofersen ASO, AAV silencing) is disease-modifying, lowering NfL and, in some patients, arresting progression ([PMID: 41661214](#), [PMID: 41670738](#)). Variant identity is the dominant prognostic modifier (A4V ~1.4 yr vs recessive D90A slow), reflecting differences in protein destabilization and aggregate strain propensity ([PMID: 18055113](#), [PMID: 40450581](#)).

Evidence Base

PMID	Contribution
41661214	Long-term tofersen (VALOR); SOD1 = ~2% of ALS; tofersen first/only approved SOD1-ALS therapy
18055113	A4V founder allele, ~1.4 yr survival; SOD1 ~20% of familial ALS; no linked modifier
36385230	D90A = most common worldwide; normal enzymatic activity → proves gain-of-function; dorsal-column pathology
41850233	Tofersen lowers plasma neurofilament (PD biomarker)
41670738	Tofersen → chronic nonprogressive ALS phenotype
42698373	NEFL most robust biomarker; proteomic risk model AUC 0.72
42661170	Registry survival: median 2.5 yr from onset; 12% ≥10 yr; predictors
42335888	Disease-associated motor neuron (DM) state; conserved mouse↔human
41702846	Misfolding gain-of-function; prion-like seeding in vivo
32958236	Aggregation pathology proportions (TDP-43 97% / SOD1 ~2% / FUS ~1%); prion-like mechanism
40364643	Four FDA-approved drugs; AMX0035 withdrawn
42265995	Juvenile SOD1 (p.Asp125Gly); lower-limb LMN onset; incomplete penetrance
26515627	Genetics ~20% (10% familial); environmental risk factor list
42669596	Military service predicts poorer ALS survival
42618698	Gold Coast criteria sensitivity 0.96
38870925	Non-uniform worldwide ALS distribution
40328546	Respiratory involvement cardinal terminal feature
25613506	Non-cell-autonomous mechanism (microglia/astrocytes/muscle/T cells)
39257530	AAV9 SOD1 suppression in neurons+astrocytes improves survival
37382103	Presymptomatic NfL; first ALS prevention trial (ATLAS)
42173382	Tofersen mechanism (mutant SOD1 mRNA degradation)
40450581	WT SOD1 coaggregation; aggregate strains; D90A recessivity

PMID	Contribution
42406382	Tofersen CNS distribution; 45–84% SOD1 reduction in human autopsy
42666355	Precision medicine, platform trials, translational gaps

Limitations and Knowledge Gaps

- SOD1 is a small minority of ALS (~2%).** Findings from SOD1 models may not generalize to the 97% TDP-43-driven majority; the SOD1 mouse does not model TDP-43 proteinopathy (PMID: [42666355](#)).
- Some epidemiologic figures** (incidence/prevalence, sex ratio, mean onset age) are drawn from established background knowledge rather than variant-specific SOD1 registries; SOD1-specific incidence is not precisely quantified here.
- One citation**
([P 37382103](#))
was flagged as a snippet mismatch during verification; the presymptomatic-NfL/ATLAS claim should be re-verified against the primary abstract before database ingestion.
- Penetrance estimates are qualitative** ("incomplete, age-dependent"); quantitative age-specific penetrance curves per variant are not established here.
- Canine degenerative myelopathy / SOD1** and detailed OMIA cross-species data were not independently verified within this investigation's citation set.
- No original data analysis** was performed; the report synthesizes published literature (no primary datasets were provided).

Proposed Follow-up Actions

- Verify flagged citation**

([P 37382103](#))
and source quantitative ATLAS design details and interim results from ClinicalTrials.gov (NCT04856982).

2. **Retrieve variant-specific penetrance and survival tables** from ClinVar/ClinGen and large SOD1 cohorts to populate genotype–phenotype annotations.
 3. **Confirm cross-species natural disease** (canine degenerative myelopathy *SOD1*) via OMIA and primary literature; add NCBI Gene IDs for orthologs.
 4. **Extract precise SOD1-ALS epidemiology** (population-specific carrier frequencies of A4V and D90A) from gnomAD and founder-population studies.
 5. **Curate ontology mappings** (HPO frequency data, GO/CL/UBERON/NCIT/CHEBI IDs) into structured fields for knowledge-base ingestion.
 6. **Track HEALEY platform and next-generation AAV/gene-editing programs** for updated therapeutic annotations.
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Report compiled from 5 investigation iterations, 14 confirmed findings, and 47 reviewed papers. Evidence types span human clinical (registries, trials, autopsy), model organism (SOD1 mouse/rat), in vitro (NSC-34, glial co-cultures), and computational/omics (snRNA-seq, proteomics).

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